ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)
JAIPUR CENTER
Academic Session : 2021-22

SRGBT

for
JEE (MAIN+ADVANCED) 2022
ENTHUSIAST COURSE

Date : 15-08-2021 | Time : 3.00 Hr | CLASS-XII

Student's Form No. : ..... **Batch :** .....

Student's Name : .....

INSTRUCTIONS

1. **PART-1 : Physics** : **(Q. No. 1 to 5)** : **Single Correct Type Questions : Marking Scheme (+3, –1)**
   (Q. No. 6 to 13) : **One or More Correct Type Questions : Marking Scheme (+4, –2)**
   Partial marks (+1) : (For darkening a bubble corresponding to each correct option, provided no incorrect option darkened)
   (Q. No. 14 to 18) : **Numerical Type Questions : Marking Scheme (+4, 0)**

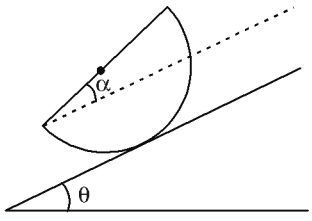
2. **PART-2 : Chemistry** : **(Q. No. 19 to 23)** : **Single Correct Type Questions : Marking Scheme (+3, –1)**
   (Q. No. 24 to 31) : **One or More Correct Type Questions : Marking Scheme (+4, –2)**
   Partial marks (+1) : (For darkening a bubble corresponding to each correct option, provided no incorrect option darkened)
   (Q. No. 32 to 36) : **Numerical Type Questions : Marking Scheme (+4, 0)**

3. **PART-3 : Mathematics** : **(Q. No. 37 to 41)** : **Single Correct Type Questions : Marking Scheme (+3, –1)**
   (Q. No. 42 to 49) : **One or More Correct Type Questions : Marking Scheme (+4, –2)**
   Partial marks (+1) : (For darkening a bubble corresponding to each correct option, provided no incorrect option darkened)
   (Q. No. 50 to 54) : **Numerical Type Questions : Marking Scheme (+4, 0)**

PART-1 : PHYSICS

SINGLE CORRECT TYPE (+3, –1)

1. A uniform thin hemispherical shell is kept static on an inclined plane of angle $\theta = 30°$ as shown. If the surface of the inclined plane is sufficiently rough to prevent sliding then the angle α made by the plane of hemisphere with inclined plane is :



(A) value of μ is needed (B) 30°

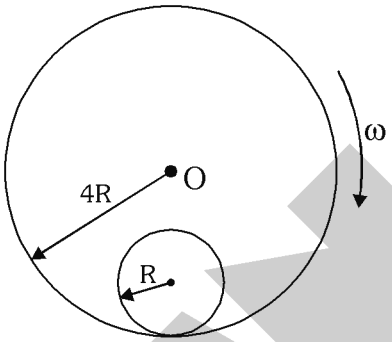
(C) 45° (D) 60°

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2. A block of mass 2.0 kg is sliding on a frictionless horizontal floor with a velocity of 5.0 m/s due east. It is subjected to a variable net force for 10 sec. resulting in an impulse of 6.0 N.s in the north and 18.0 N.s in the west. What is the total work done by this force ?

(A) 0.0 J (B) 90 J (C) 100 J (D) 200 J

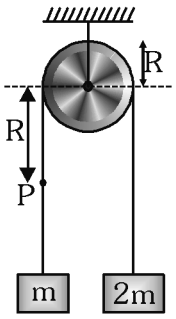
3. In a gravity free space, a hollow cylinder of radius 4R and mass M is being rotated by a constant angular veloicty ω clockwise w.r.t an inertial frame of reference. Another hollow cylinder of radius R and mass m is rotating inside this cylinder, such that it touches the inner surface of larger cylinder without slipping and its centre of mass appears stationary from inertial frame Dynamic friction coefficient between cylinders is μ. If the normal reaction between cylinders is denoted by N.



A small tangential velocity v is given to the smaller cylinder towards right. What will be N, once the steady state is achieved. (Outer cylinder is rotating about fixed axis passing through center O and perpendicular to the plane of paper).

(A) $m\frac{v^2}{16R}$ (B) $m\frac{v^2}{12R}$ (C) $m\frac{(3R\cdot(3\omega/4)+v)^2}{3R}$ (D) zero

4. The given system is released from rest then find change in acceleration of particle 'P' in the interval it just comes in contact with the pulley to just before it leaves the pulley. Pulley and string are massless but sufficiently rough :



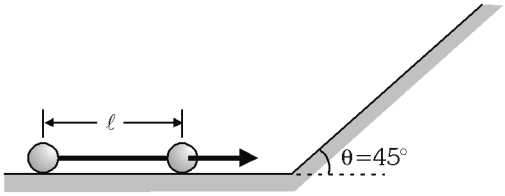
(A) $\frac{2g}{3}\sqrt{1+(2+\pi)^2}$

(B) $\frac{g}{3}\sqrt{1+(2+\pi)^2}$

(C) $\frac{g}{3}\sqrt{1+(1+\pi)^2}$

(D) $\frac{g}{3}\sqrt{1+(1+\pi)^2}$

5. Two identical particles connected with a massless rod are placed on a frictionless horizontal floor, which terminates into a uniform frictionless slope of inclination θ = 45°. Find minimum velocity given to the particles towards the slope so that the rod can rise completely on the slope. Collisions of the particle with the slope are perfectly inelastic.



(A) $\sqrt{\frac{g\ell}{2\sqrt{2}}}$ (B) $\sqrt{\frac{\sqrt{2}g\ell}{3}}$ (C) $\sqrt{\frac{2\sqrt{2}g\ell}{3}}$ (D) $\sqrt{\frac{3g\ell}{2\sqrt{2}}}$

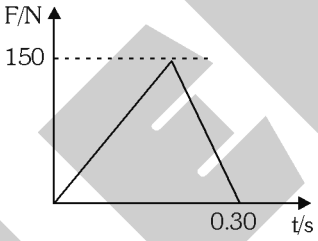
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ONE OR MORE THAN ONE CORRECT TYPE (+4, –2)

● Answer to each question will be evaluated according to the following marking scheme:

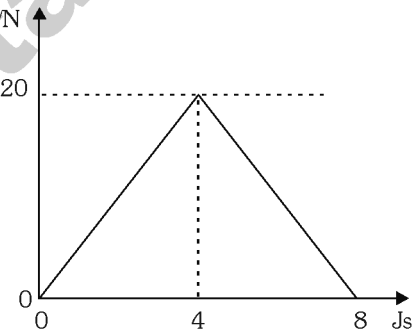
Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

6. Two bodies A and B of masses 5.00 kg and 10.0 kg moving in free space in opposite directions with velocities 4.00 m/s and 0.50 m/s respectively undergo a head-on collision. The force F of their mutual interaction varies with time t according to the given graph. What can you conclude from the given information ?



(A) Period of deformation is 0.20 s.
(B) Coefficient of restitution is 0.5.
(C) Body A will move with velocity 0.5 m/s in the original direction.
(D) Body B will move with velocity 1.75

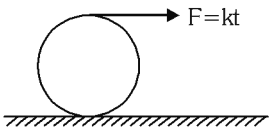
7. A box of mass 4 kg placed on a horizontal floor, is acted upon by a horizontal force F that varies with time t as shown in the given graph. If coefficients of static and kinetic frictions are 0.25, on which of the following conclusions can you arrive ?



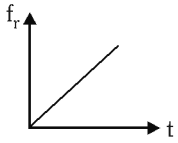
(A) The box starts moving at the instant t = 2.0 s.
(B) The maximum velocity acquired by the box is 5.0 m/s
(C) The box stops at the instant t = 9.0 s.
(D) Modulus of average power of the force F is more than that of the frictional force.

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8. A uniform solid cylinder of mass m, radius R is at rest on an extremely rough horizontal surface. Now a horizontal force F = kt where k = constant and t = time, is applied at the highest point on the cylinder. Assume that the cylinder is not slidding at all.



(A) The friction force acting on the cylinder varies with time as

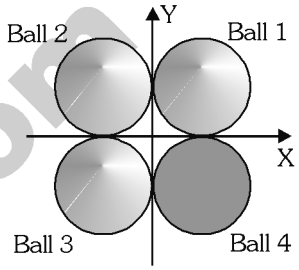


(B) Velocity of the highest point after time t will be $\frac{2kt^2}{3m}$

(C) Speed of the particle which is at highest point of the cylinder after time t will be $\frac{4kt^2}{3m}$

(D) If the coefficient of friction between the ground and the cylinder is μ, the cylinder will start slidding at $t = \frac{3\mu mg}{k}$

9. 4 identical balls of radius R and mass m are lying in a gravity free space. The balls are in contact and there centres are forming vertices of a square of side 2R in horizontal plane. One identical ball travelling vertically with a speed v hits the four balls symmetrically. The collision is perfectly elastic. The centre of the 4 stationary ball are at (R, R, 0), (R, –R, 0), (R, –R, 0) and (–R, –R, 0) initially.



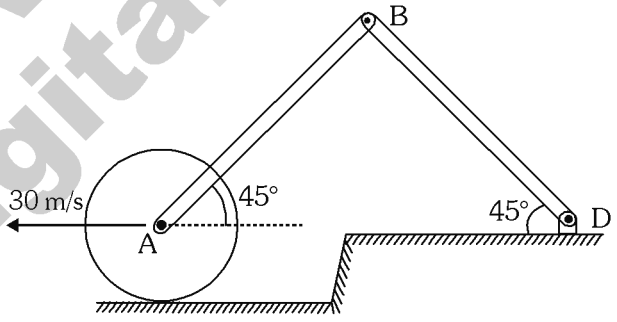
(A) The speed of ball 1 after collision is $\frac{v}{\sqrt{2}}$

(B) The speed of ball 1 after collision is $\frac{\sqrt{2}}{3}v$

(C) The speed of ball which was initially travelling with speed v just after collision is v/3

(D) The speed of ball which was initially travelling with speed v just after collision is $v/\sqrt{3}$

10. A 60 mm radius wheel is connected to a fixed support D by two links AB and BD. At the instant shown, the velocity of the centre A of the wheel is 30 m/s to the left. Rod AB and BD have equal length ℓ = 5m. Then at the given instant :



(A) The angular velocity of rod AB will be $3\sqrt{2}$ rad / sec

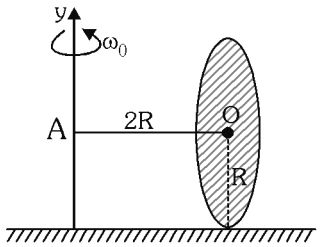
(B) The angular velocity of rod BD is $3\sqrt{2}$ rad / sec

(C) Velocity of pin B will be $5\sqrt{2}$ m / s

(D) Velocity of pin B will be $15\sqrt{2}$ m / s

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11. The given uniform disc is rotating about vertical y-axis with constant angular speed ω_0. If surface is sufficiently rough then (OA is a massless rod and plane of disc is perpendicular to the rod



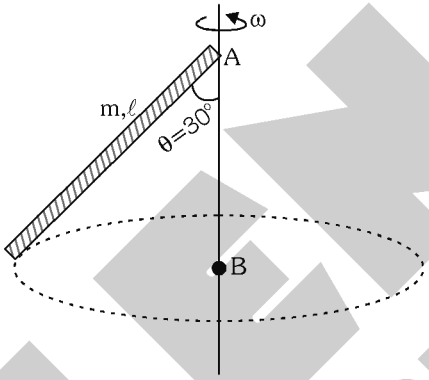
(A) angular velocity of disc about an axis passing through the centre O and perpendicular to the disc is $2\omega_0$.

(B) Kinetic energy of the disc is $\frac{25}{8}mR^2\omega_0^2$

(C) Acceleration of point of contact with ground is $2\sqrt{5}\omega_0^2R$

(D) Kinetic energy in the reference frame of center of mass 'O' is $\frac{9}{8}mR^2\omega_0^2$

12. The given uniform rod is hinged at point A and rotates with constant angular velocity ω about vertical axes AB.



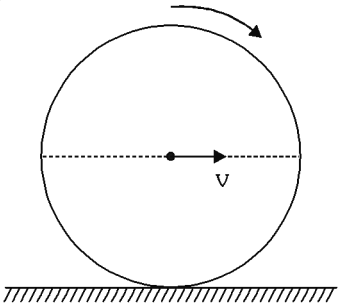
(A) Angular momentum of the rod about point A is $\frac{m\ell^2\omega}{12}$

(B) Angular momentum of the rod about point A is $\frac{m\ell^2\omega}{6}$

(C) Angular momentum of the rod about axis AB is $\frac{m\ell^2\omega}{12}$

(D) Angular momentum of the rod and torque about point A are always perpendicular to each other.

13. The given hollow sphere of mass M and radius R is in pure rolling with constant linear velocity $v = \sqrt{gR}$. Then at the given instant :



(A) Force exerted by the upper hemisphere on the lower hemisphere is zero

(B) Force exerted by the upper hemisphere on the lower hemisphere is Mg/4

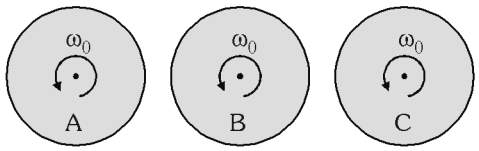
(C) Kinetic energy of lower hemisphere is $\frac{1}{3}Mv^2$

(D) Kinetic energy of lower hemisphere is $\frac{1}{6}Mv^2$

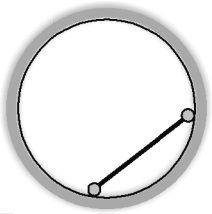
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NUMERICAL TYPE
(SINGLE/DOUBLE & THREE DIGITS ANSWER INCLUDING DECIMAL) (+4 , 0)

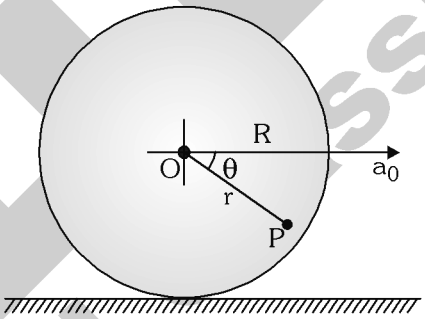
14. Three identical uniform cylinders are mounted on frictionless horizontal axles passing through their centres. They are given equal angular velocities ω_0 each and then gradually brought into contact preventing any vertical movement and rotation of their axles. The angular velocities of the cylinders after they stop slipping is $\omega_0 x$ then find x ?



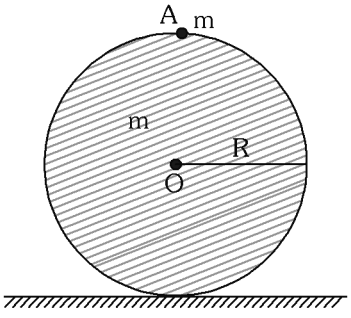
15. Inside a fixed frictionless spherical cavity of radius r, a dumbbell is held touching the bottom of the cavity at its lower end. The dumbbell consists of two identical small balls each of mass m affixed at the ends of a light rod of length $r\sqrt{2}$. The force of interaction between the lower ball and the cavity be immediately after the dumbbell is released is x mg then find x.



16. The wheel of radius R rolls without slipping on horizontal rough surface, and its centre O has an horizontal acceleration a_0 in forward direction. A point P on the wheel is a distance r from O and angular position θ from horizontal. For a given values of a_0, R and r, the angle θ for which point P has no acceleration in this position is $\sin^{-1}\left(\frac{xr}{yR}\right)$ then find x + y ? (x and y are lowest possible integers).

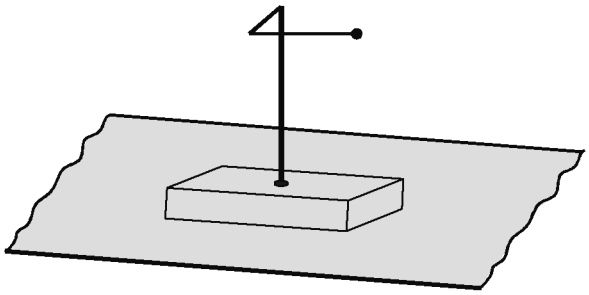


17. The given disc is slightly rotated and then released. The angular acceleration of the system after the particle 'A' has rotated by angle 90° about centre O is $\frac{xg}{yR}$ then find x + y ? The surface is sufficiently rough. (x and y are lowest possible integers).



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18. A block of mass m can slide without friction on a horizontal floor. a small ball of same mass m is suspended by a light inextensible cord of length ℓ from a hanger fixed at the top of a vertical pole mounted on the block so that the ball can swing freely in a vertical plane. The ball is released from a horizontal position as shown. The maximum tensile force in the string during the subsequent motion is x mg then find x ?



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PART-2 : CHEMISTRY

SINGLE CORRECT TYPE (+3, –1)

19. Translational, rotational and vibrational degrees of freedom contribute to the molar heat capacity at constant volume, C_V. For $AX_{3(g)}$, C_V is 18 cal/deg and for $M_2H_{2(g)}$ C_V is 19 cal deg^{-1}, one can therefore conclude that :

(A) Both AX_3 and M_2H_2 are linear
(B) Both AX_3 and M_2H_2 are non-linear
(C) AX_3 is linear and M_2H_2 is non-linear
(D) AX_3 is non-linear and M_2H_2 is linear

20. Which of the following option is incorrect ?

(A) The difference between ionisation enthalpy (ΔH_{IE}) and ionisation energy (IE) is directly depencent on absolute temperature.
(B) heat added to the system at lower temperature causes greater randomness as compared to same amount of heat added at higher temperature.
(C) A container having N_2O_4 gas if heated will show an increase in intensity of brown colour.
(D) It is easier to remove H^+ from HPO_4^{2-} as compared to that from H_2PO_4.

21. From the following data.

$$C(s, graphite) + O_2(g) \longrightarrow CO_2(g) \qquad \Delta H^\circ_{reaction} = -393.5\,kJ/mole$$

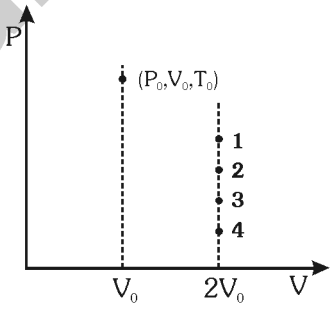
$$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(\ell) \qquad \Delta H^\circ_{reaction} = -286\,kJ/mole$$

$$2C_2H_6(g) + 7O_2(g) \longrightarrow 4CO_2(g) + 6H_2O(\ell) \qquad \Delta H^\circ_{reaction} = -3120\,kJ/mole$$

Calculate the standard enthalpy of formation of C_2H_6 (g) (in kJ/mole).
(A) –85 (B) –170 (C) 170 (D) 85

22. If four identical samples of an ideal gas initially at similar state (P_0, V_0, T_0) are allowed to expand to double their volumes byy four different process :

(I) by isothermal irrevrsible process.
(II) by reversible process such that P^2V = constant
(III) by reversible adiabatic process.
(IV) by irreversible adiabatic expansion against constnt external pressure.
Then in the graph shown the final state is represented by four different points then, the correct match can be :



(A) 1–(I), 2–(II), 3–(III), 4–(IV)
(B) 1–(II), 2–(I), 3–(IV), 4–(III)
(C) 2–(III), 3–(II), 4–(I), 1–(IV)
(D) 3–(II), 1–(I), 2–(IV), 4–(III)

23. Which of the following options represents correct sequence of True(T) or False(F) statements.

Statement - I No Gibbs free energy change occurs four vapourisation at triple point.
Statement - II Entropy driven processes are spontaneous at high temperature.
Statement - III ΔG° of a reaction will increase with increase in temperature if the reaction is endothermic.
(A) T T T (B) T T F (C) F T F (D) T F F

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ONE OR MORE THAN ONE CORRECT TYPE (+4, –2)

● Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

24. Mention the correct statement(s) among the following :

(A) Keeping initial states same, final temperature for reversible adiabatic expansion is more than irreversible adiabatic expansion if final volume is also same.

(B) $A(s) \rightleftharpoons A(g)$ $\quad \log_{10} = 4 - \dfrac{2000}{T}$ $\quad \Delta H_{Sub}$ for A (s) is 9.212 kCal/mole.

(C) For 1 mole of N_2 gas $\quad \underset{(0°C,\,1atm)}{\text{State A}} \xrightarrow[\text{Irreversible}]{\text{Single step}} \underset{(0°C,\,2\,atm)}{\text{State B}}$ $\quad Q_{Process} = -22.4$ L-atm if $P_{ext} = 2$ atm

(D) For 1 mole of O_2 gas $\quad \underset{(0°C,\,1atm)}{\text{State A}} \xrightarrow[\text{Irreversible}]{\text{Single step}} \underset{(0°C,\,2\,atm)}{\text{State B}}$ $\quad \Delta H = 0$ and $W_{process} = 44.8$ L-atm if $P_{ext} = 2$ atm

25. One mole of an ideal gas ($C_{V,m} = \dfrac{5}{2}R$) at 300 K and 5 atm is expanded adiabatically against constant atmospheric pressure to a final pressure of 2 atm. Then which is/are true.

(A) Final temperature of the gas is approx 251.4 K.

(B) Final temperature of the gas is approx 270 K.

(C) $P_1V_1^{\gamma} = P_2V_2^{\gamma}$ is valid for the process.

(D) If the process is carried out in infinite steps then final temperature will be less than the temperature observed in above process.

26. If an equation like $N_2O_4(g) \rightleftharpoons 2NO_2(g)$, the ΔG (reaction) expression given is

$$\Delta G = 4.73 \text{ kJ mol}^{-1} + 2.5 \text{ kJ mol}^{-1}\ \ell n\left\{\frac{P_{NO_2}^2}{P_{N_2O_4}}\right\} \text{ then}$$

(A) If small amount of N_2O_4 is introduced in a closed container, then reaction moves in forward direction spontaneously.

(B) Initially, $\Delta G°$ value is negative for the given decomposition.

(C) At equilibrium, $Q_P = K_P = \text{antilog}\left|\dfrac{-\Delta G°}{RT}\right|$

(D) At equilibrium state ΔG must be zero.

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27. Select the correct option(s) :

(A) $\Delta H_f^\circ[H_2O(\ell)]$ is zero at 25°C

(B) $\Delta H_f^\circ[O_2(g)]$ is zero at 25°

(C) $S^\circ[O_2(g)]$ is zero at 25°C

(D) Entropies of aqueous ions can be negative or positive.

28. Which of the following statement are true?

(A) The entropy of the universe increases and tends towards the maximum value

(B) All natural processes are irreversible

(C) Entropy of the system is maximum when the system is at equilibrium

(D) According to the third law of thermodynamics, $S_{0K}^0 = 0$ for any element or compound.

29. 1 mole of a real gas changes its state from – A (2 bar, 3L, 100 K) to state – B (2 bar, 5 L, 200 K) at constant pressure and finally to state – C (3 bar, 10 L, 300 K). If $\Delta U_{BC} = 110$ J and $C_{p,m}$ of gas = 3 R = 3 × 8.3 JK^{-1} mol^{-1} then choose the correct option(s) :

(A) $W_{AB} = 830$ J (B) $\Delta H_{AC} = 4600$ J (C) $\Delta U_{AC} = 2200$ J (D) $\Delta U_{AC} = 1770$ J

30. Assume ideal gas behaviour for all the gases considered & vibrational degree of freedom to be active separate equimolar samples of He, H_2, SO_2 & CH_4 were subjected to a two step process as mentioned. Initially all are at same state of temperature & pressure.

Step I → All undergo reversible adiabatic expansion to attain same final volume, which is double the original volume thereby causing the decreases in their temperature.

Step II → After step I all are given appropriate amount of heat isochorically to restore the original temperature.

Mark the correct option(s).

(A) Due to step I only, the decreases in temperature will be maximum for CH_4.

(B) During step II, heat given will be maximum for CH_4.

(C) There will be no change in intermal energy for any of the gas after both the steps of process are completed.

(D) The P–V graph of CH_4 & SO_2 will be same.

31. Which is correct relation between various enthalpies given all enthalpies are measured under similar pressure and temperature conditions-

(A) $2\Delta_f H^\circ(HCl,g) = BE_{(H-H)} + BE_{(Cl-Cl)} - 2BE_{(H-Cl)}$

(B) $\Delta_f H^\circ(CO_2,g) = \Delta H^\circ_{comb.}$ (C, graphite)

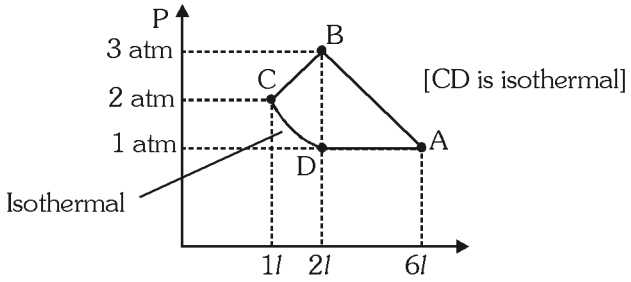
(C) ΔH° for conversion of d-lactic acid to *l*-lactic acid is zero

(D) $\Delta H^\circ_{neutralization}$ (HCl + NaOH) is equal to ΔH° for the reaction $H^+(aq.) + OH^-(aq.) \rightarrow H_2O(l)$

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NUMERICAL TYPE
(SINGLE/DOUBLE & THREE DIGITS ANSWER INCLUDING DECIMAL) (+4 , 0)

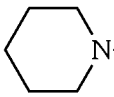
32. One mole of a monoatomic ideal gas is subjected to a process as shown. Calculate the maximum temperature obtained during the process (in Kelvin)

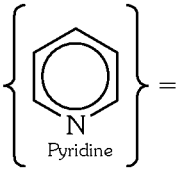


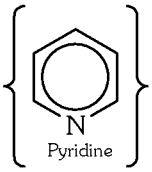
[If $T_{max} = \frac{X}{R}$, then fill X in the OMR sheet]

33. Calculate $\Delta H_{\text{resonance energy}}$ of pyridine.

Given data : ΔH_f° pyridine$_{(\ell)}$ = –175 kJ/mole

Compound	$\Delta H^\circ_{f,298K}$ **in kJ/mole**
$CH_2 = N - CH_3(g)$	+44
$CH_3 - NH - CH_3(g)$	–18
 (g)	–37
 (g)	–156
 N—H (g)	–50

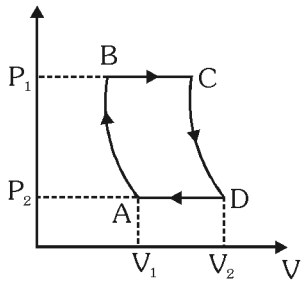
Resonance energy  = –125 kJ/mole

$\Delta H_{vap\ ,\ 298\ K}$  = +40 kJ/mole

34. Certain quantity of an ideal gas is expanded from 20 L to 60 L isothermally in three steps. Pressure at these three stages are 1.5 atm, 1.2 atm and 1 atm respectively now the gas is brought back to the initial condition by these three stages again. Calculate the total work done on the system during this cycle in joule.
(Take : 1 atm – L = 100 J)

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35. An ideal gas is carried through a thermodynamic cycle ABCD.



$P_2 = 10$ bar $\qquad V_1 = \dfrac{1}{\ell n10}$ litre

$P_1 = 1$ bar $\qquad V_2 = \dfrac{4}{\ell n10}$ litre

Consisting of two isobaric and two isothermal process as shown in figure. Calculate the magnitude of net work in the entire cycle in litre bar.

36. A reversible cyclic process involves 6 steps. In step -1, 3 system absorb 500 J, 800 J of heat from a heat reservoir at temperature 250K & 200K respectively. Step 2, 4, 6, are adiabatic such that the temperature of one reservoir changes to that of next. Total work done by the system in whole cycle is 700 J. Find the temperature during step 5 if it exchange heat from a reservoir at temperature T_5

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PART-3 : MATHEMATICS

SINGLE CORRECT TYPE (+3, –1)

37. Circles are drawn through the points (a, b) and (b, –a) such that the chord joining the two points subtends an angle of 45° at any point of the circumference. The the distance between the centres is :

(A) $\sqrt{3}$ times the radius of either circle

(B) 2 times the radius of either circle

(C) $\frac{1}{\sqrt{2}}$ times the radius of either circle

(D) $\sqrt{2}$ times the radius of either circle

38. If four points P, Q, R, S in the plane be taken and the square of the length of the tangent from P to the circle on QR as diameter be denoted {P, QR}, then :

(A) {P, RS} – {P, QS} + {Q, PR} – {Q, RS} = 0

(B) {P, RS} + {P, QS} + {Q, PR} + {Q, RS} = 0

(C) {P, RS} – {P, QS} + {Q, PR} – {Q, RS} = 1

(D) {P, RS} – {Q, RS} – {Q, PR} – {P, QS} = 0

39. A variable triangle ABC is circumscribed about a fixed circle of unit radius, Side BC always touches the circle at fixed point D. If B and C vary in such a way that BD.DC = 2, then the locus of vertex A will lie on straight line:

(A) parallel to side BC

(B) perpendicular to side BC

(C) Making an angle of $\frac{\pi}{6}$ with side BC

(D) Making an angle of $\sin^{-1}\left(\frac{2}{3}\right)$ with side BC

40. Let A and B be two fixed points and P, another point in the plane, moves in such a way that $k_1PA + k_2PB = k_3$, where k_1, k_2 and k_3 are real constants. Then which of the following is locus of P :

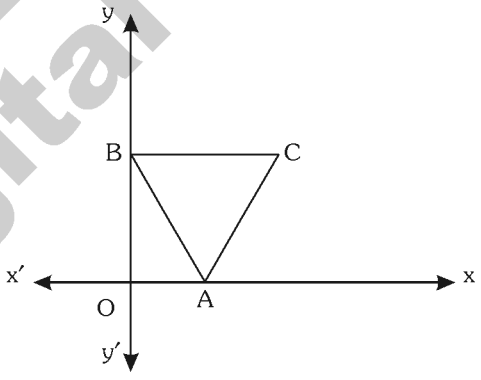
(A) an ellipse, if $k_2 = -1$ and $k_1, k_3 > 0$

(B) a circle, if $k_1 > 0$, $k_2 < 0$, $k_3 = 0$

(C) an ellipse, if $k_1 = k_2 > 0$ and $k_3 > 0$

(D) a circle, if $k_1 = 0$ and $k_2, k_3 > 0$

41. adjacent figure represents a equilateral triangle ABC of side length 2 units. Locus of vertex 'C' ass the side AB slides along the coordinates axes is :



(A) $x^2+y^2-xy+1=0$

(B) $x^2+y^2+xy\sqrt{3}=1$

(C) $x^2+y^2=1+xy\sqrt{3}$

(D) $x^2+y^2-xy\sqrt{3}+1=0$

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ONE OR MORE THAN ONE CORRECT TYPE (+4, –2)

● Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

42. Which of the following statementss is true about circles $x^2+y^2+2x=0$ and $x^2+y^2-6x=0$:

(A) length of the transverse common tangents is $\sqrt{2}$ units

(B) length of the direct common tangents is $2\sqrt{3}$ units

(C) length of the transverse common tangents is 0 units

(D) area of trapezium, enclosed by the common tangents and chords of contact is $6\sqrt{3}$ units^2

43. A line segment of length 7 units has one of its ends on x-axis and other end on y-axis. A point P divides this line segment in the ratio 3 : 4 locus of P will be an ellipse with :

(A) length of latus rectum $\frac{9}{2}$ (B) eccentricity $\frac{\sqrt{7}}{4}$

(C) length of major axis 8 (D) None of these

44. A circle is touching the lines $x - 2y = 3$ and $2x + y = 1$, the locus of its centre is :

(A) $3x - y - 4 = 0$ (B) $x - 3y - 4 = 0$ (C) $x + 3y + 2 = 0$ (D) $3x + y - 2 = 0$

45. If the line $x + 2y = 3$ is rotated about its point of intersection with the line $x = y$ by an angle of 45°, then it's equations in the new position is/are :

(A) $3x - y - 2 = 0$ (B) $x + 3y - 4 = 0$ (C) $3x + y - 4 = 0$ (D) $x - 3y + 2 = 0$

46. A circle $S_2 = 0$ has its centre lying on the circle $S_1 : x^2 + y^2 = 4$ and radius half of that of S_1. If a common tangent of these circles is inclined at an angle of 30° with positive direction of x-axis, then possible equation(s) of S_2 is/ are?

(A) $x^2+y^2+4x+3=0$ (B) $x^2+y^2-4x+3=0$

(C) $x^2+y^2-2x-2\sqrt{3}y+3=0$ (D) $x^2+y^2+2x+2\sqrt{3}y+3=0$

47. Cosider the points O(0, 0), A(0, 1), B(1, 1) in the xy plane, suppose that points C(x, 1) and D(1, y) on a line passing through O are chosen such that $0 < x < 1$. Let sum of areas of DAC and DBCD be 'S' then which of the following are correct?

(A) min. value of S is irrational in $\left(\frac{1}{3}, \frac{1}{2}\right)$

(B) min. value of S is irrational in $\left(\frac{2}{3}, 1\right)$

(C) The value of x for minimum value of S lies in $\left(\frac{2}{3}, 1\right)$

(D) The value of x for minimum values of S lies in $\left(\frac{1}{3}, \frac{1}{2}\right)$

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48. A tangent drawn from the point (4, 0) to circle $x^2 + y^2 = 8$ touches it at a point A in the first quadrant. The coordinates of another point B on the circle such that $\ell(AB) = 4$:

(A) (2, –2) (B) (–2, 2) (C) (–2, –2) (D) (2, 2)

49. Two circles of equal radius, having centers at (2, 3) and (5, 6) cut each other orthogonally. Their common chord will pass through the point(s)

(A) (2, 6) (B) (6, 4) (C) (7, 1) (D) (3, 5)

NUMERICAL TYPE
(SINGLE/DOUBLE & THREE DIGITS ANSWER INCLUDING DECIMAL) (+4 , 0)

50. The vertices of a triangle is (4, 3), (–4, 3) and (5, 0). The orthocenter of the triangle is (a, b). The $|a - b|$ is equal to :

51. A ray along line $x + 2y - 3 = 0$ is reflected by mirror $x = y$, if reflected line has equation $ax + by - 3 = 0$. Find $a+b$.

52. In the triangle ABC coordinates of A are (1, 2), Bisector of internal angle B is $2x - y = 5$ and perpendicular bisector of the side AC is $3x - 4y = 20$. If equation of BC is $ax + by = c$, then $\frac{c-a}{b}$=?

53. The chord of contact of tangents from A to the circle $x^2 + y^2 = a^2$ passes through a point B. If 3 and 4 are the length of tangents from A and B on the given circle then AB equals to :

54. P is a point inside the triangle ABC. Lines are drawn through P, parallel to the sides of the triangle. The three resulting triangles with the vertex at P have areas 4, 9 and 49 sq. units. The area of the triangle ABC is

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SRGBT-4

ENTHUSIAST COURSE

DATE : 15-08-2021

ANSWER KEY

PART-1 : PHYSICS

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	A	B	A	D	ABD	ABC	ACD	BC	ABD
Que.	11	12	13	14	15	16	17	18		
Ans.	ABCD	BCD	BD	0.33	1.5	2	67	5		

PART-2 : CHEMISTRY

Que.	19	20	21	22	23	24	25	26	27	28
Ans.	D	D	A	B	B	BC	BD	ACD	BD	ABD
Que.	29	30	31	32	33	34	35	36		
Ans.	BC	BC	ABCD	8	85	3500	3	100		

PART-3 : MATHEMATICS

Que.	37	38	39	40	41	42	43	44	45	46
Ans.	D	A	A	D	C	BCD	ABC	AC	CD	ABCD
Que.	47	48	49	50	51	52	53	54		
Ans.	AC	AB	ACD	1	3	12	5	144		